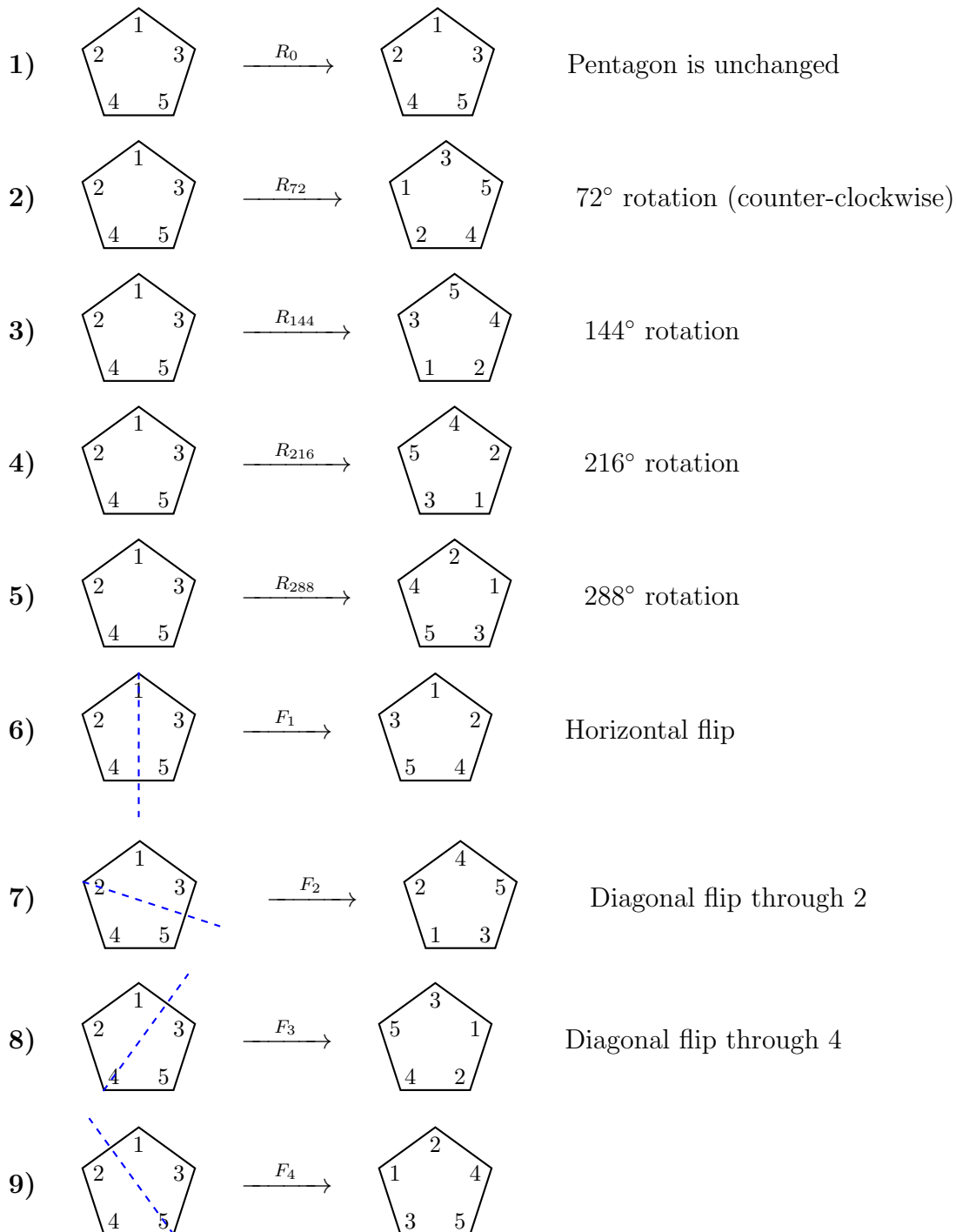


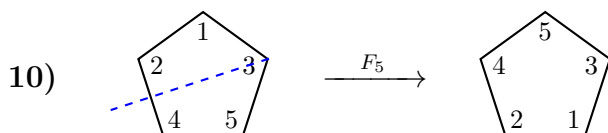
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1. Find and describe the elements of D_5 , the dihedral group of order 10. Specify the order of each element, and construct a Cayley table for the group.

* $D_5 \Rightarrow$ pentagon

* 5 sides $\Rightarrow$ Every rotation should be a multiple of $360 \div 5 = 72^\circ$





Cayley Table for D_5

	R_0	R_{72}	R_{144}	R_{216}	R_{288}	F_1	F_2	F_3	F_4	F_5
R_0	R_0	R_{72}	R_{144}	R_{216}	R_{288}	F_1	F_2	F_3	F_4	F_5
R_{72}	R_{72}	R_{144}	R_{216}	R_{288}	R_0	F_3	F_4	F_5	F_1	F_2
R_{144}	R_{144}	R_{216}	R_{288}	R_0	R_{72}	F_5	F_1	F_2	F_3	F_4
R_{216}	R_{216}	R_{288}	R_0	R_{72}	R_{144}	F_2	F_3	F_4	F_5	F_1
R_{288}	R_{288}	R_0	R_{72}	R_{144}	R_{216}	F_4	F_5	F_1	F_2	F_3
F_1	F_1	F_4	F_2	F_5	F_3	R_0	R_{144}	R_{288}	R_{72}	R_{216}
F_2	F_2	F_5	F_3	F_1	F_4	R_{216}	R_0	R_{144}	R_{288}	R_{72}
F_3	F_3	F_1	F_4	F_2	F_5	R_{72}	R_{216}	R_0	R_{144}	R_{288}
F_4	F_4	F_2	F_5	F_3	F_1	R_{288}	R_{72}	R_{216}	R_0	R_{144}
F_5	F_5	F_3	F_1	F_4	F_2	R_{144}	R_{288}	R_{72}	R_{216}	R_0

$$\begin{aligned}
 |R_0| &= 1 \text{ (identity)} \\
 |R_{72}| &= 5 \text{ (} R_{72}R_{72}R_{72}R_{72}R_{72} = R_0 \text{)} \\
 |R_{144}| &= 4 \text{ (} R_{144}R_{144}R_{144}R_{144} = R_0 \text{)} \\
 |R_{216}| &= 3 \text{ (} R_{216}R_{216}R_{216} = R_0 \text{)} \\
 |R_{288}| &= 2 \text{ (} R_{288}R_{288} = R_0 \text{)} \\
 |F_1| &= 2 \text{ (} F_1F_1 = R_0 \text{)} \\
 |F_2| &= 2 \text{ (} F_2F_2 = R_0 \text{)} \\
 |F_3| &= 2 \text{ (} F_3F_3 = R_0 \text{)} \\
 |F_4| &= 2 \text{ (} F_4F_4 = R_0 \text{)} \\
 |F_5| &= 2 \text{ (} F_5F_5 = R_0 \text{)}
 \end{aligned}$$

2. Prove that the set of all 3×3 matrices with real entries of the form

$$\begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$$

is a group under matrix multiplication. That is,

$$\begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & a+a' & b'+ac'+b \\ 0 & 1 & c'+c \\ 0 & 0 & 1 \end{bmatrix}$$

To prove that this set is a group, we must show that it satisfies all the properties of a group.

Associativity: Yes, because matrix multiplication is inherently associative.

Identity:

$$\begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \boxed{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

Inverse:

$$\begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \boxed{\begin{bmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix}}$$

$\Rightarrow M_{3 \times 3}$ of this form is a group

3. Prove that a group G is Abelian if and only if $(ab)^{-1} = a^{-1}b^{-1}$ for all a and b in G .

Proof. Suppose G is Abelian and $ab \in G$ s.t. $\exists (ab)^{-1}$. Using the Socks-Shoes Property,

$$(ab)^{-1} = b^{-1}a^{-1}.$$

Since G is Abelian, commutativity is satisfied and thus,

$$\begin{aligned} b^{-1}a^{-1} &= a^{-1}b^{-1} \\ \Rightarrow (ab)^{-1} &= a^{-1}b^{-1}. \end{aligned}$$

Suppose $(ab)^{-1} = a^{-1}b^{-1}$. Since $a, b \in G$, using the Socks-Shoes Property,

$$\begin{aligned} (ab)^{-1} &= b^{-1}a^{-1} \\ \Rightarrow a^{-1}b^{-1} &= b^{-1}a^{-1} \\ \Rightarrow (ba)^{-1} &= (ab)^{-1}. \end{aligned}$$

Since inverses are unique, it must be that

$$ba = ab$$

which is precisely the definition of commutativity, and $\therefore G$ must also be Abelian. $\square$

4. Prove that if G is a group with the property that the square of every element is the identity, then G is Abelian.

Proof. Suppose $G = \{x \mid x^2 = e\}$ is a group. Let $a, b \in G$ s.t

$$\begin{aligned} a^2 &= b^2 = e \\ \Rightarrow a^2 b^2 &= ee = e \\ \Rightarrow b^2 a^2 &= ee = e \\ \Rightarrow a^2 b^2 &= b^2 a^2. \end{aligned}$$

Since G is a group, $a^{-1}, b^{-1} \in G$. Multiplying a^{-1} on the LHS and b^{-1} on the RHS of both expressions,

$$a^{-1}(a^2 b^2)b^{-1} = a^{-1}(b^2 a^2)b^{-1}.$$

Using the property of associativity and exponent rules,

$$\begin{aligned} a^{-1+2}b^{2-1} &= a^{-1}(ee)b^{-1} \\ \Rightarrow ab &= a^{-1}b^{-1}. \end{aligned}$$

By the Socks-Shoes Property,

$$\begin{aligned} ab &= (ba)^{-1} \\ \Rightarrow ab &= (ba)^{-1}e \\ \Rightarrow ab &= (ba)^{-1}(ba)^2 \text{ where } ba \in G \\ \Rightarrow ab &= (ba)^{-1+2} \\ \Rightarrow ab &= ba. \end{aligned}$$

This is the definition of commutativity on elements of G showing that G is Abelian. □

5. List the six elements of $GL(2, \mathbb{Z}_2)$. Show that this group is non-Abelian by finding two

elements that do not commute.

- * $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$
- $\Rightarrow \mathbb{Z}_2 = \{0, 1\}$
- * Elements of a GL group have nonzero determinants

$$GL(2, \mathbb{Z}_2) = \mathbf{1)} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (\det = 1 \neq 0)$$

$$\mathbf{2)} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (\det = -1 \neq 0)$$

$$\mathbf{3)} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad (\det = -1 \neq 0)$$

$$\mathbf{4)} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad (\det = 1 \neq 0)$$

$$\mathbf{5)} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad (\det = 1 \neq 0)$$

$$\mathbf{6)} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \quad (\det = -1 \neq 0)$$

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\Rightarrow \boxed{AB \neq BA, \therefore GL(2, \mathbb{Z}_2) \text{ is non-Abelian.}}$$

6. Let a belong to a group G with $|a| = m$. If n is relatively prime to m , show that a can be written as the n th power of some element in the group.

Proof. Suppose G is a group in which $a \in G$ and a is of order m . Letting b be some element in G , we want to show that

$$b^n = a.$$

If n is relatively prime with m , that means

$$\gcd(n, m) = 1.$$

By Bezout's Identity, $\exists x, y \in \mathbb{Z}$ s.t

$$\begin{aligned} xn + ym &= 1 \\ \Rightarrow a^{xn+ym} &= a^1. \end{aligned}$$

Using exponent rules,

$$\begin{aligned} a^{xn} \cdot a^{ym} &= a \\ \Rightarrow (a^x)^n \cdot (a^m)^y &= a. \end{aligned}$$

Since the order of a is m , $a^m = e$. Thus,

$$\begin{aligned} (a^x)^n \cdot e^y &= a \\ \Rightarrow (a^x)^n &= a \\ \Rightarrow b^n &= a \text{ where } b = a^x \in G. \end{aligned}$$

□